

The allegory axioms, and what defines them in the Frobenius calculus

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

- write F as $/$, G as $\backslash$

- render $a \leq b$ as $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

- **rewrite** $X \leq XF$ as $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\begin{aligned} \boxed{\frac{X/}{X/}} &\xRightarrow{\{\text{adj}\}} \boxed{\frac{X}{X\backslash}} \xRightarrow{\{\text{rewrite}\}} \boxed{\frac{*}{\bigwedge}} \xRightarrow[\{\text{put } / \text{ on right}\}]{\{\text{put } \backslash \text{ on left}\}} \boxed{\frac{\backslash}{\bigwedge}} \\ &\xRightarrow[\{\text{put } / \text{ on right}\}]{\{\text{put } \backslash \text{ on left}\}} \boxed{\frac{/}{\bigvee}} \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\{\text{adj}\}} \boxed{\frac{Y\bigvee}{Y}} \xRightarrow{\{\text{rewrite}\}} \boxed{\frac{\bigvee}{*}} \xRightarrow[\{\text{put } \backslash \text{ on right}\}]{\{\text{put } / \text{ on left}\}} \boxed{\frac{\bigvee}{/}} \\ &\xRightarrow[\{\text{put } \backslash \text{ on right}\}]{\{\text{put } / \text{ on left}\}} \boxed{\frac{\bigwedge}{\backslash}} \end{aligned}$$

now we have $\bigvee = /$ and $\bigwedge = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

$F \dashv G$	F 's type	monad FG	$1 \leq FG$	$GF \leq 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangle$	$A \rightarrow A \otimes A$	$\triangleleft \triangle = 1$	$1 \leq \triangleleft \triangle$	$\triangle \triangle \leq 1$	$(RuS) \triangleleft = R \triangleleft uS \triangleleft$	$(RnS) \triangle = R \triangle nS \triangle$	$\triangle \triangle \triangle = \triangle$	$\triangle \triangle \triangle = \triangle$
$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \leq \multimap \multimap$	$\multimap \multimap \leq 1$	$(RuS) \multimap = R \multimap uS \multimap$	$(RnS) \multimap = R \multimap nS \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$\circ \circ = 1$	$R = R \circ \circ$	$R = R \circ \circ$	$(RuS) \circ = R \circ uS \circ$	$(RnS) \circ = R \circ nS \circ$	$R \circ \circ = R \circ$	$R \circ \circ = R \circ$
$\multimap \triangle \dashv \triangle \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\multimap \triangle \otimes 1) (1 \otimes \multimap \triangle) = 1$	$(1 \otimes \multimap \triangle) (\triangle \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangle \multimap \dashv \multimap \triangle$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \multimap \triangle) (\triangle \otimes 1) = 1$	$(\multimap \triangle \otimes 1) (1 \otimes \multimap \triangle) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv n$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \mapsto RnR = R$	$R \leq RnR$	$RnS \leq R$	$\Delta(RuS) = \Delta Ru \Delta S$	$(RnT)n(SnU) = (RnS)n(TnU)$	$RnR = R$	$RnR = R$
$u \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (RuS, RuS)$	$R \leq RuS$	$RuR \leq R$	$(RuT)u(SuU) = (RuS)u(TuU)$	$\Delta(RnS) = \Delta Rn \Delta S$	$RuR = R$	$RuR = R$
$\perp \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$	—	—	$\perp \leq R$	—	—	—	—
$\cdot S \dashv /S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \leq (RS)/S$	$(T/S)S \leq T$	$(RuT)S = RSuTS$	$(RnT)/S = R/SnT/S$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \leq S \setminus (SR)$	$S(S \setminus T) \leq T$	$S(RuT) = SRuST$	$S \setminus (RnT) = S \setminus RnS \setminus T$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$Rn \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (XnR)$	$X \leq R \Rightarrow (XnR)$	$Rn(R \Rightarrow Y) \leq Y$	$Rn(XuY) = (RnX)u(RnY)$	$R \Rightarrow (XnY) = (R \Rightarrow X)n(R \Rightarrow Y)$	$Rn(R \Rightarrow (XnR)) = XnR$	$R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \leq (\mathcal{D}R)T$	$\mathcal{D}(AT) \leq A$	$\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$	$(AnB)T = ATnBT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \leq T(\mathcal{R}R)$	$\mathcal{R}(TA) \leq A$	$\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$	$T(AnB) = TAnTB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$1 \leq ff^\circ$	$f^\circ f \leq 1$	$(RuS)f = RfuSf$	$(RnS)f^\circ = Rf^\circ nSf^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$1 \leq ff^\circ$	$f^\circ f \leq 1$	$f^\circ(XuY) = f^\circ Xu f^\circ Y$	$f(XnY) = fXn fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv E$	$\text{Map} \hookrightarrow \text{Rel}$	P	$\{\cdot\} : A \rightarrow P \ A$	$\exists : P \ B \rightarrow B$	—	—	$\Lambda(\exists) = 1$	$\Lambda(R)\exists = R$
$\cdot \exists \dashv \Lambda$	$\text{Map}(A, P \ B) \rightarrow (A \rightarrow B)$	1	$\Lambda(f\exists) = f$	$\Lambda(R)\exists = R$	—	—	$\Lambda(f\exists)\exists = f\exists$	$\Lambda(\Lambda(R)\exists) = \Lambda(R)$
$\Lambda \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, P \ B)$	1	$\Lambda(R)\exists = R$	$\Lambda(f\exists) = f$	—	—	$\Lambda(\Lambda(R)\exists) = \Lambda(R)$	$\Lambda(f\exists)\exists = f\exists$

(1.1a)

§1.2. Composing adjunctions

One more adjunction first, which §1.1 cannot hold because it is ANTITONE — its left adjoint turns joins into meets, so that table's F preserves u column would read the wrong way; its type is contravariant, $F : (B \rightarrow C)^{\text{op}} \rightarrow (A \rightarrow B)$, so the F 's type column cannot be written in §1.1's form either; and BOTH composites are inflationary, $1 \leq FG$ and $1 \leq GF$, so the $GF \leq 1$ column has no entry for it at all. That last is the signature of an antitone Galois connection — two closure operators rather than a closure and an interior — and the table's own **both units** cell is it:

$F \dashv G$	the law	both units
$R/ \dashv \backslash R$	$T \leq R/S$ iff $S \leq T \backslash R$	$S \leq (R/S) \backslash R$, $T \leq R/(T \backslash R)$

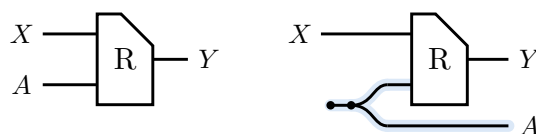
(1.2a)

Two rows of §1.1 composed are a third adjunction, and its right adjoint read in the two orders is a law. Row first, column second; right adjoints compose the other way round, which is where every reversal in the table comes from.

The table's $\multimap \triangle$ row and column need a name for that row's left adjoint, and §1.1 gave it none: write $R^\wedge$ for the operator listed there, $R \mapsto (1 \otimes \multimap \triangle)(R \otimes 1) : X \rightarrow Y \otimes A$.

$$R : X \otimes A \rightarrow Y \quad R^\wedge = (1 \otimes \multimap \triangle)(R \otimes 1) : X \rightarrow Y \otimes A$$

(1.2b)



(1.2c)

left R , right $R^\wedge$: the pale strand is everything the bend adds, and the box and the X, Y wires are where they were.

In $\mathbf{Rel}$ the two are one set of triples split two ways — the bend moves the A coordinate from the input side to the output side:

$$x \ R^\wedge \ (y, a) \iff (x, a) \ R \ y \quad (1.2d)$$

Read $\mathbf{Rel}(I, J)$ as an $I \times J$ boolean matrix — Freyd’s §2.111, which defines reciprocation entrywise as $j \ R^\circ \ i = i \ R \ j$ — and $R^\wedge$ is the smallest of three degrees of bending, all of the same $R : X \otimes A \rightarrow Y$:

	what moves	type	as a matrix	(1.2e)
$R^\wedge$	one index, to the output side	$X \rightarrow Y \otimes A$	rows re-indexed from (x, a) to x , columns from y to (y, a)	
R°	the input index goes out and the output index comes in drawn in the $^\circ$ section below	$Y \rightarrow X \otimes A$	the transpose	
$\lceil R \rceil$	everything, to the output side — the name of R	$I \rightarrow Y \otimes A \otimes X$	the whole matrix flattened into one row	

All three carry the same entries; only the split of the indices into row and column changes. That is what a matrix transpose always was — moving an index from the domain side to the codomain side — and compact closure is the structure that permits the move: $\mathbf{Rel}$ is self-dual, $A^* = A$, so there is no up/down index distinction to keep track of.

Of the three, only $^\wedge$ is this note’s own, the half bend having no settled symbol. $\lceil R \rceil$ and its mirror the **coname** $\lfloor R \rfloor : Y \otimes A \otimes X \rightarrow I$ are the literature’s. Neither is the transpose Λ of B&dM, which is Freyd’s §2.421 $R/S = \Lambda(R)\Lambda^\circ(S)$ as well: that one sends $R : X \otimes A \rightarrow Y$ to a MAP $\Lambda(R) : X \otimes A \rightarrow P \ Y$, and needs a power allegory rather than compact closure.

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\circ \triangleleft$	Δ	(1.2f)
$\cdot S$	$R/(ST) = (R/T)/S$	$T \backslash (R/S) = (T \backslash R)/S$	$R/(Sg) = (Rg^\circ)/S$	$g(R/S) = (gR)/S$	—	$(Y/S)^\circ = S^\circ \backslash (Y^\circ)$	$(RS)^\wedge = R^\wedge(S \otimes 1)$	$(T_1 n T_2)/S = T_1/S n T_2/S$	
$S \cdot$	$S \backslash (R/T) = (S \backslash R)/T$	$(TS) \backslash R = S \backslash (T \backslash R)$	$S \backslash (Rg^\circ) = (S \backslash R)g^\circ$	$(g^\circ S) \backslash R = S \backslash (gR)$	—	$(S \backslash Y)^\circ = Y^\circ/S^\circ$	$((S \otimes 1)R)^\wedge = S \ R^\wedge$	$S \backslash (T_1 n T_2) = S \backslash T_1 n S \backslash T_2$	
$\cdot f$	$R/(fT) = (R/T)f^\circ$	$T \backslash (Rf^\circ) = (T \backslash R)f^\circ$	$(fg)^\circ = g^\circ f^\circ$	—	—	$(fY)^\circ = Y^\circ f^\circ$	—	$(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$	
$f^\circ \cdot$	$f(R/T) = (fR)/T$	$(Tf^\circ) \backslash R = f(T \backslash R)$	—	$(fg)^\circ = g^\circ f^\circ$	—	$(fY)^\circ = Y^\circ f^\circ$	—	$f(T_1 n T_2) = fT_1 n fT_2$	
Rn	—	—	—	—	$(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$	
$^\circ$	$(Y/T)^\circ = T^\circ \backslash (Y^\circ)$	$(T \backslash Y)^\circ = Y^\circ/T^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$Y^\circ \Rightarrow Y$	$(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section’s display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$	
$\circ \triangleleft$	$(RT)^\wedge = R^\wedge(T \otimes 1)$	$((T \otimes 1)R)^\wedge = T \ R^\wedge$	—	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section’s display	—	—	
u	$(X_1 u X_2)T = X_1 T u X_2 T$	$T(X_1 u X_2) = T X_1 u T X_2$	$(X_1 u X_2)g = X_1 g u X_2 g$	$g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$	$Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$	$(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$	—	—	
$\perp$	$\perp T = \perp$	$T \perp = \perp$	$\perp g = \perp$	$g^\circ \perp = \perp$	$Tn \perp = \perp$	$\perp^\circ = \perp$	—	—	

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists f \dashv f^* \dashv \forall f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

$$\begin{aligned} \cdot f &\dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot &\dashv f \cdot \dashv f \backslash \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1’s division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \backslash$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $R/(f \ S) = (R/S) \ f^\circ$. — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\mathbf{Rel}(I, A) = P \ A$ they are the image triple:

operator	acts by	name
$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

(1.3b)

§1.1's $\mathcal{D} \dashv \cdot \tau$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = P(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, $A\tau$ is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R/\tau = \{(a, a) : \forall b. aRb\}$.

$$\mathcal{D} \dashv \cdot \tau \dashv 1 \cap \cdot / \tau$$

(1.3c)

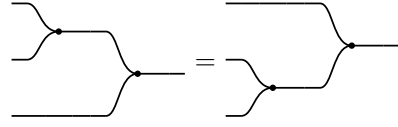
Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\forall_f \perp = B \setminus \text{im } f$ is still not $\perp$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^{\circ\circ} = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv \cdot \dashv \cdot \dashv \cdot \dashv \cdot$ has no end, and §1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.

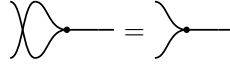
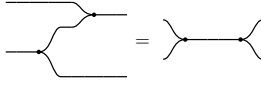
§2. Relations

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ \dashv \otimes$, and

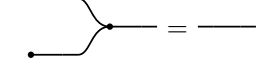
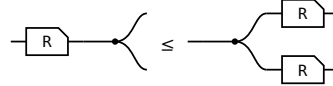
$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \circ : \mathbb{I} \rightarrow A)$ a commutative monoid with $\mathbf{C}^\circ \dashv \mathbf{C}$. We write $\mathbf{C}^\circ$ as $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow \mathbb{I})$



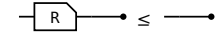
$\triangleright$ associative



$\triangleright$ commutative



$\circ$ is its unit



Frobenius, one half — the other is its $\circ$

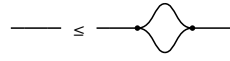
$R\triangleleft \leq \triangleleft(R\otimes R)$

$R\circ \leq \circ$

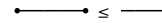
§2.1. $\forall$ object $\mathbf{A}$, $(\mathbf{A}, \triangleleft, \circ) \dashv (\mathbf{A}, \triangleright, \circ)$



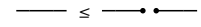
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft \triangleright$



$\circ \circ \leq 1$



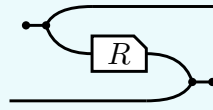
$1 \leq \circ \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$^\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

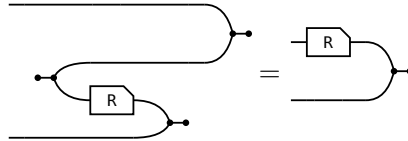
where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\circlearrowright$ are $\triangleright^\circ$ and $\circlearrowleft^\circ$. The laws of $^\circ$ are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)

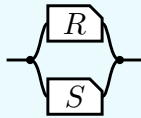


R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

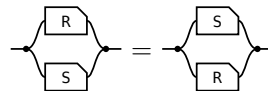
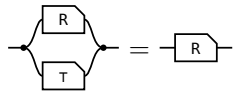
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



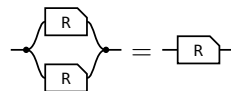
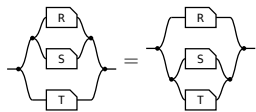
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).

(4b)



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

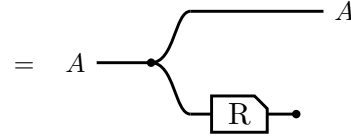
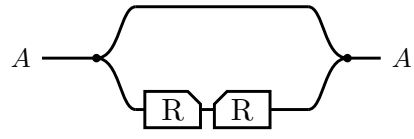
And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Domain and range

The **domain** $\text{Dom } R \triangleq 1 \cap R R^\circ$ and the **range** $\text{Ran } R \triangleq \text{Dom } R^\circ$.



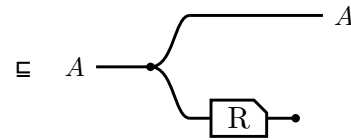
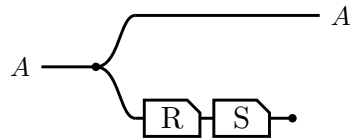
$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{Ran } R$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{Dom } R \triangleq 1 \cap R R^\circ$
$\text{Dom } R \subseteq A \iff R \subseteq A R, \text{ for } A \text{ coreflexive}$
$\text{Dom } (R S) \subseteq \text{Dom } R$
$\text{Dom } (R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Dom } R = 1 \iff 1 \subseteq R R^\circ$
$R \text{ simple} \iff R^\circ R \subseteq 1$
$R \text{ a map} \iff R \text{ entire and simple}$
$R, S \text{ entire} \implies R S \text{ entire} \text{ — likewise simple, likewise maps}$
$R S \text{ entire} \implies R \text{ entire}$

§5.1. Sliding the discard

$\text{Dom } (R S) \subseteq \text{Dom } R$, and a single glyph for Dom would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

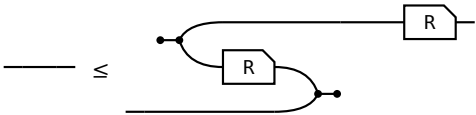
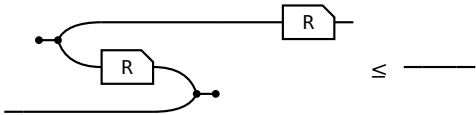
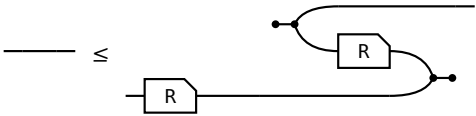
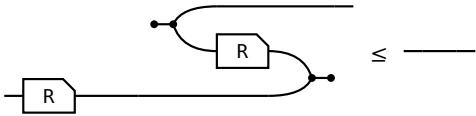


$S \circ \subseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

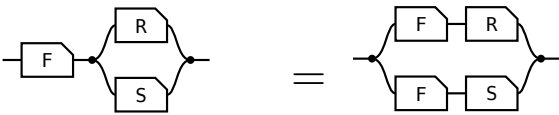
Equality is $S \text{ entire}$, which is the same picture read as $\text{Dom } R = 1 \iff R \text{ entire}$.

§6. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by taking $\circ$ of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

 surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	 single valued $R^\circ R \sqsubseteq 1$
 entire $1 \sqsubseteq R R^\circ$. With single valued, a map.	 injective $R R^\circ \sqsubseteq 1$

(6a)

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

(6b)

§7. / is all of

(7a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S\backslash R) y \iff \forall p. p S x \rightarrow p R y$$

/ compares images: $S(y) \subseteq R(x)$. $\backslash$ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

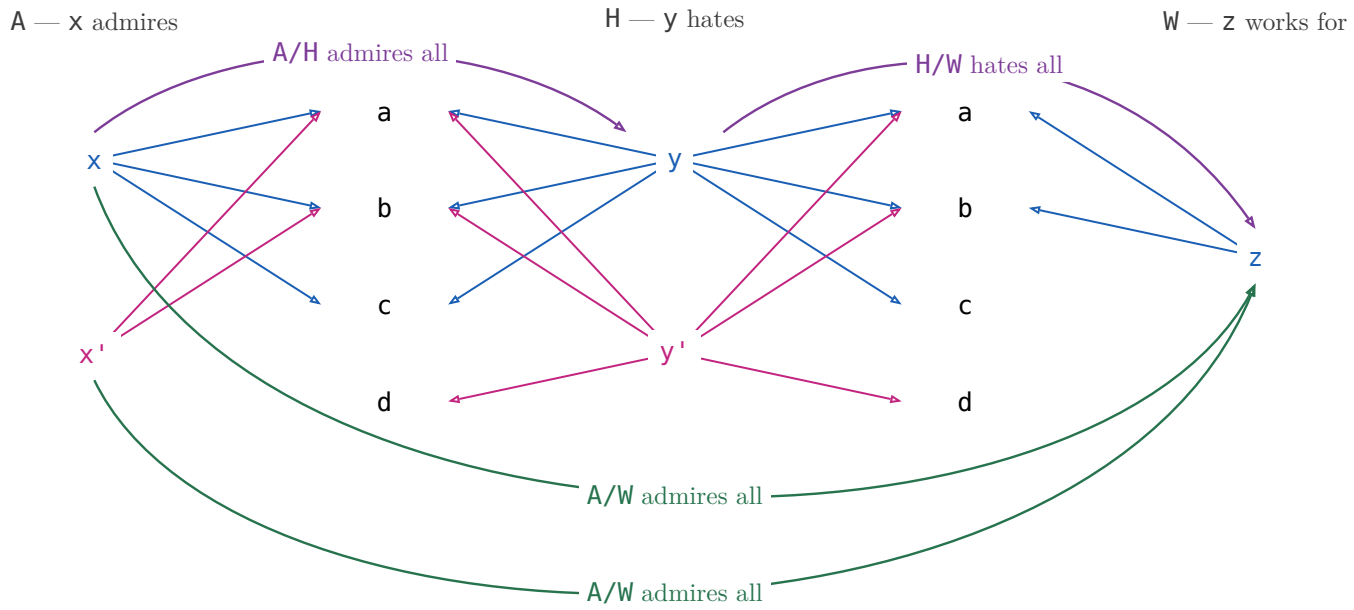
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H\backslash A) y$ — everyone who hates x admires y .

§7.1. $(R/S)(S/W) \sqsubseteq R/W$

(7.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(7.1b)

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

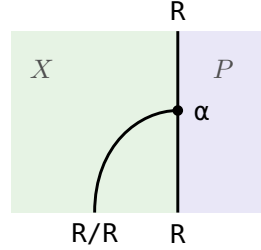
$X \sqsubseteq R/S \iff X S \sqsubseteq R$ X is any X -to- Y pairing; one that only pairs X with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \sqsubseteq S \setminus R \iff S X \sqsubseteq R$ The mirror — divide on the left when X comes first.	
$(R/S) S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \emptyset$: R/S is everyone, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f (R/S) = (f R)/S$ Rename x before or after dividing — the licence to write $f R / S$.	
$R/(f S) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S) (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-set that already covers everyone Z works for admires those people too.	
$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
$(R/R) R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§7.2. R/R is a preorder, which is what a monad is here

A **monad** in a locally posetal 2-category is a 1-cell M with $1 \sqsubseteq M$ and $M M \sqsubseteq M$ — a unit and a multiplication, and with the hom-posets thin there is nothing else to give. In $\mathbf{Rel}$ that is a reflexive transitive relation: a preorder. The table above proves both halves at $M := R/R$, and names the preorder they define.

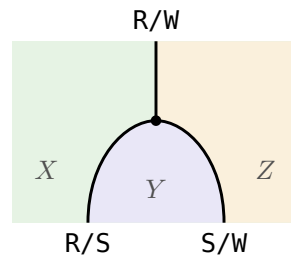
$(R/R) R = R$ is then the action that makes R a module over that monad — Hinze and Marsden’s $\alpha : M \circ A \rightarrow A$ (IntroString p. 83) at $M := R/R$, $A := R$. Their two coherence conditions are free here: parallel 2-cells are unique, so any two of them are equal.



(7.2a)

Green is X , where x lives; purple is P , where the p of the definition lives. $R/R : X \rightarrow X$ runs inside X and ends on $R : X \rightarrow P$ at the action α , which is $(R/R) R \sqsubseteq R$.

§7.3. The whole family is one preorder



(7.3a)

The same dot with one region more: $R/S : X \rightarrow Y$ and $S/W : Y \rightarrow Z$ come in, $R/W : X \rightarrow Z$ goes out. Green X , purple Y , amber Z .

Closing Y off is the mid-point y being quantified away; the $\sqsubseteq$ rather than $=$ is the loss the figure for that law above already draws.

Concretely in $\mathbf{Rel}$, $1 \sqsubseteq R/R$ and $(R/S)(S/W) \sqsubseteq R/W$ are one preorder on the disjoint union of $X, Y, Z, \dots$: reflexivity is $1 \sqsubseteq R/R$ inside a block, transitivity is $(R/S)(S/W) \sqsubseteq R/W$ across blocks.

Abstractly the same two are a **lax functor** out of the codiscrete category on $\{R, S, W, \dots\}$ — one object per relation, exactly one arrow each way — sending R to X , the object its rows are indexed by, and the arrow $R \rightarrow S$ to $R/S : X \rightarrow Y$. That is Bénabou’s **polyad** (Bénabou 1967, Def. 5.5), of which a monad is the one-object case: the subsection above is this one at $R = S = W$.

R/S is at the same time a bimodule between the two monads R/R and S/S : $(R/R)(R/S) \sqsubseteq R/S$ and $(R/S)(S/S) \sqsubseteq R/S$, each the counit $(R/S) S \sqsubseteq R$ with one action composed on — $(R/R) R = R$ on the left, $(S/S) S = S$ on the right.

§8. $x \text{ (Admires\%Hates) } y$ is x admires only all whom y hates

$$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ. \text{ In Rel } x \text{ and } y \text{ has the same image: } \forall p. (x R p \iff y S p)$$

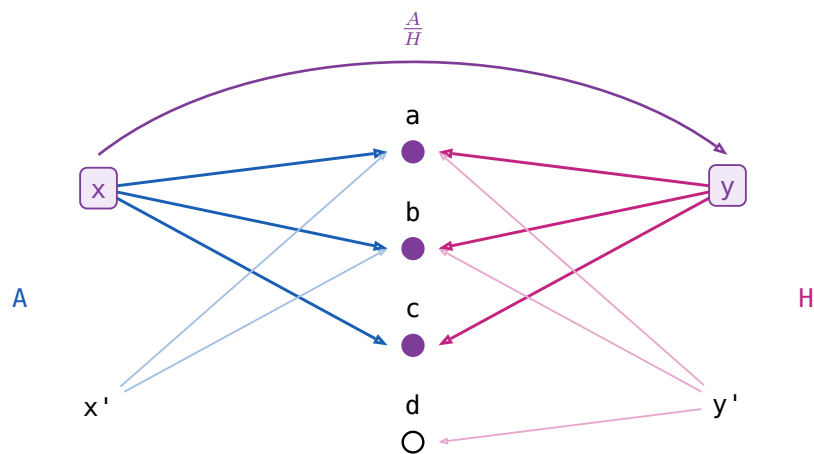
(8a)

$$\begin{aligned} x \text{ (A/H) } y &\text{ --- } x \text{ admires everyone } y \text{ hates} \\ x \text{ ((H/A)^\circ) } y &\text{ --- } x \text{ admires only people } y \text{ hates} \\ x \text{ ((A/H) \cap (H/A)^\circ) } y &\text{ --- } x \text{ admires only and all whom } y \text{ hates} \end{aligned}$$

(8b)

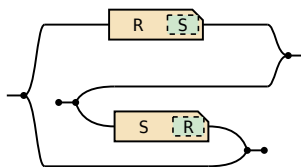
x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(8c)



A meet of two long divisions, the second turned round by the converse frame:

(8d)



exported from the definition, not transcribed

(8e)

$\left(\frac{R}{S}\right)^\circ = \frac{S}{R}$ Matching is symmetric.	$= \frac{S}{R}$
$\frac{R}{S} \frac{S}{W} \sqsubseteq \frac{R}{W}$ And transitive.	$\frac{R}{S} \frac{S}{W} \sqsubseteq \frac{R}{W}$

No pictures for the rest of §2.35: symmetric division is not built from the generators.

(8f)

law	the reading
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R \text{ and } X^\circ R \sqsubseteq S$	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$\frac{R}{S} S \sqsubseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S})R$
$\frac{R}{R} R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y = x$ always qualifies ($1 \sqsubseteq R\%R$ below)
$1 \sqsubseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(\frac{R}{R})^2 = \frac{R}{R}$	So the relation admires the same people is an equivalence relation.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.
$\frac{R}{1}$ is the simple part of R	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S)(S/R)$	Its domain is the domain of simplicity of R .

§8.1. Straight

(8.1a)

S is **straight** when $\frac{S}{S} = 1$ — no two ys hate the same people.

(8.1b)

law	the reading
every symmetric X with $X S \sqsubseteq S$ is coreflexive	Equivalently, S is straight.
$f S = g S \implies f = g$	A straight S tells its ys apart, so it cancels on the right.
$S R \text{ straight} \implies S \text{ straight}$	If the longer chain tells them apart, the first step already does.
$S \text{ straight} \iff R/S \text{ simple for all } R$	If no two ys agree, at most one x can match a given y .
$R = h S$, h a cover, S straight	In an effective division allegory every arrow factors that way.

§9. Freyd's Power allegories

(9a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, & \exists_R &= \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R &\text{ is } \mathbf{thick} \\ (\exists_R/\exists_R) \cap (\exists_R/\exists_R)^\circ &\subseteq \mathbb{1} & \exists_R &\text{ is } \mathbf{straight} \end{aligned}$$

$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : P B \rightarrow B$, dropping the subscript.

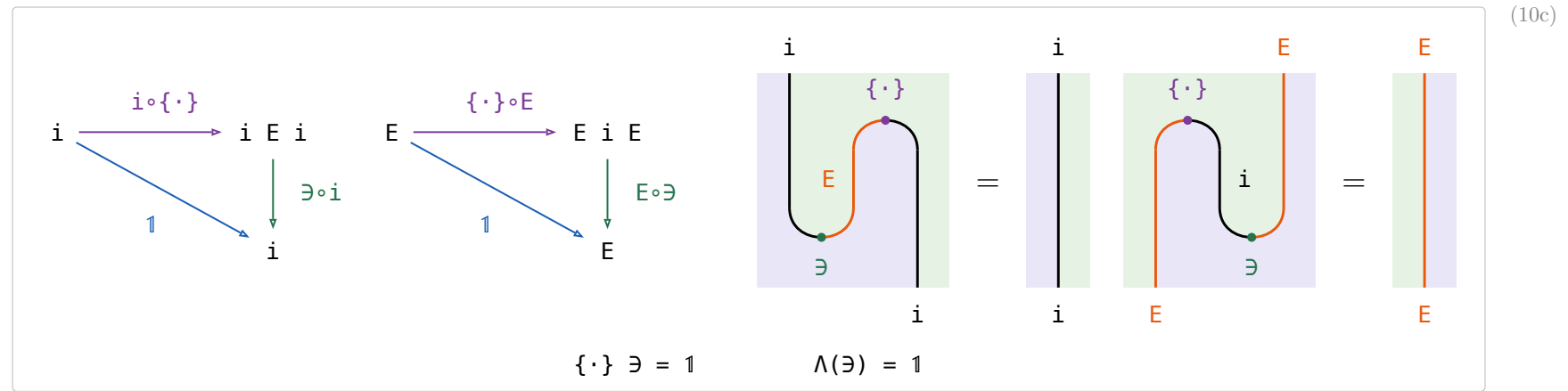
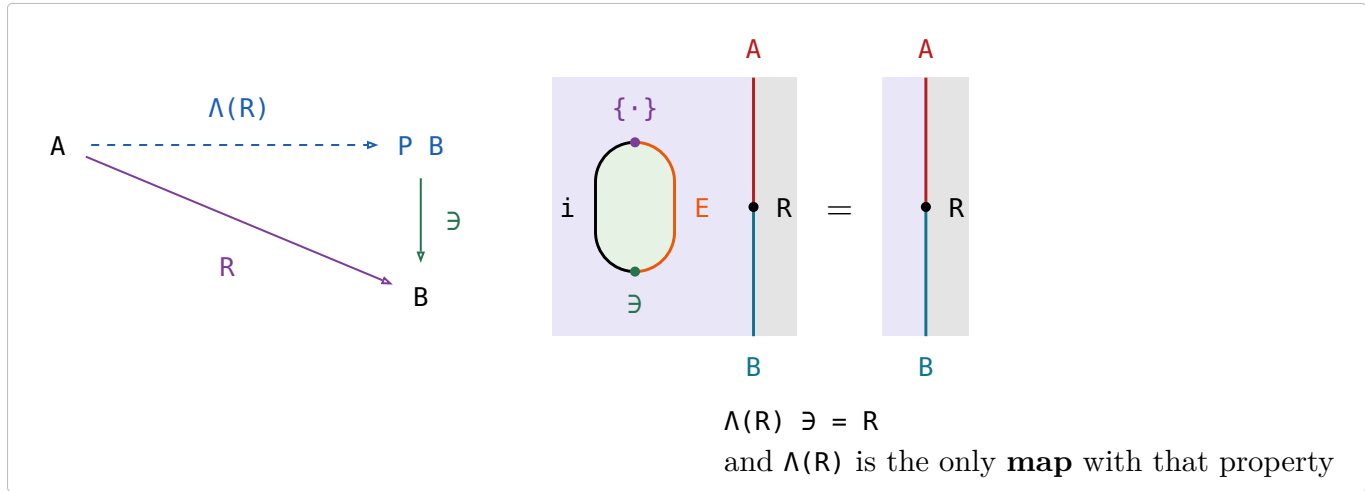
(9b)

law	the reading
$\exists_R \square = R\square, \exists_R = \exists_R \square$	$\exists$ has the same target as R , and replacing R by the identity at that target leaves it unchanged: one $\exists$ per object, not per arrow.
$\exists$ is thick	Comprehension: every x has a set of exactly the people x admires. Equivalently every R factors as a map followed by $\exists$.
$\exists$ is straight , that is $\frac{\exists}{\exists} \subseteq \mathbb{1}$	Extensionality: two sets with the same people are the same set.
$\Lambda(R) \triangleq \frac{R}{\exists} : A \rightarrow P B$, for $R : A \rightarrow B$	convert a relation to a function. $a \Lambda(R) = \{b \mid a R b\}$, image of a
$\Lambda(R)$ is simple	$\exists$ is straight, and dividing by a straight arrow is simple. At most one set per x .
$\Lambda(R)$ is entire $\iff \exists$ thick	$\text{Dom } \frac{R}{\exists} = \mathbb{1} \cap (R/\exists)(\exists/R)$, the domain row above. At least one set per x , so $\Lambda(R)$ is a map .
$\Lambda(R) \exists = R$	Drawn in §10, where it is the counit's cancellation.
$\Lambda(R)$ is the only map with $\Lambda(R) \exists = R$	Two maps naming the same people name the same set — extensionality again.
$F \subseteq \Lambda(F \exists)$, F simple	A partial choice of sets is inside the total one.
$P A \triangleq$ source of $\exists$, the power-object	Notation, not a law: $\exists$ goes $P A \rightarrow A$, and $P A$ is simply a name for where it starts. Every subset of A .
$\{\cdot\} \triangleq \Lambda(\mathbb{1})$, the singleton map , monic	The one-person set. $\Lambda(\mathbb{1})\Lambda^\circ(\mathbb{1}) \subseteq \frac{\mathbb{1}}{\exists} \frac{\exists}{\mathbb{1}} \subseteq \frac{\mathbb{1}}{\mathbb{1}} \subseteq \mathbb{1}$.
$\Lambda(\exists) = \mathbb{1}$	Make the set of a set, then read it back one level down. Straightness itself, and one of the two snake laws of §10; the other is $\{\cdot\} \exists = \mathbb{1}$.
fusion: $\Lambda(f R) = f \Lambda(R)$, f a map	Naturality of the unit, $f \{\cdot\} = \{\cdot\} (E(f))$, and that is the whole content of fusion. Drawn in §10.
$\Lambda(f) = f \{\cdot\}$, f a map	Rename first or take singletons first — the fusion row above at $R = \mathbb{1}$.
$\frac{R}{S} = \Lambda(R)\Lambda^\circ(S)$	x and y match exactly when $\Lambda(R)x$ and $\Lambda(S)y$ are one and the same set.
$E(R) \triangleq \Lambda(\exists R)$	$E(R) : P A \rightarrow P B$, image of a set of A
$\Lambda(R) = \{\cdot\} E(R)$	$\{\cdot\} : x \mapsto \{x\}$
$\Lambda(S) E(R) = \Lambda(S R)$	absorption
subset $\triangleq \exists/\exists : P A \rightarrow P A$	$x \text{ subset } y \iff \forall a. y \ni a \rightarrow x \ni a$, that is $y \subseteq x$: diagram order reverses B&dM's argument order along with the arrow. It models inclusion between sets, where $\subseteq$ is the allegory's own primitive comparing arrows (B&dM p. 106).

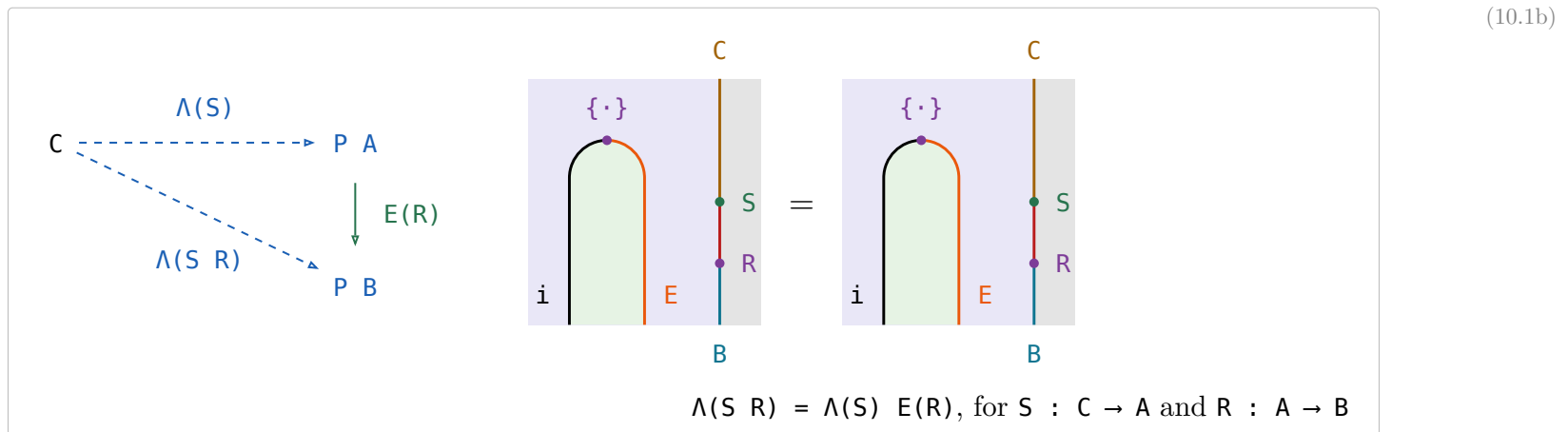
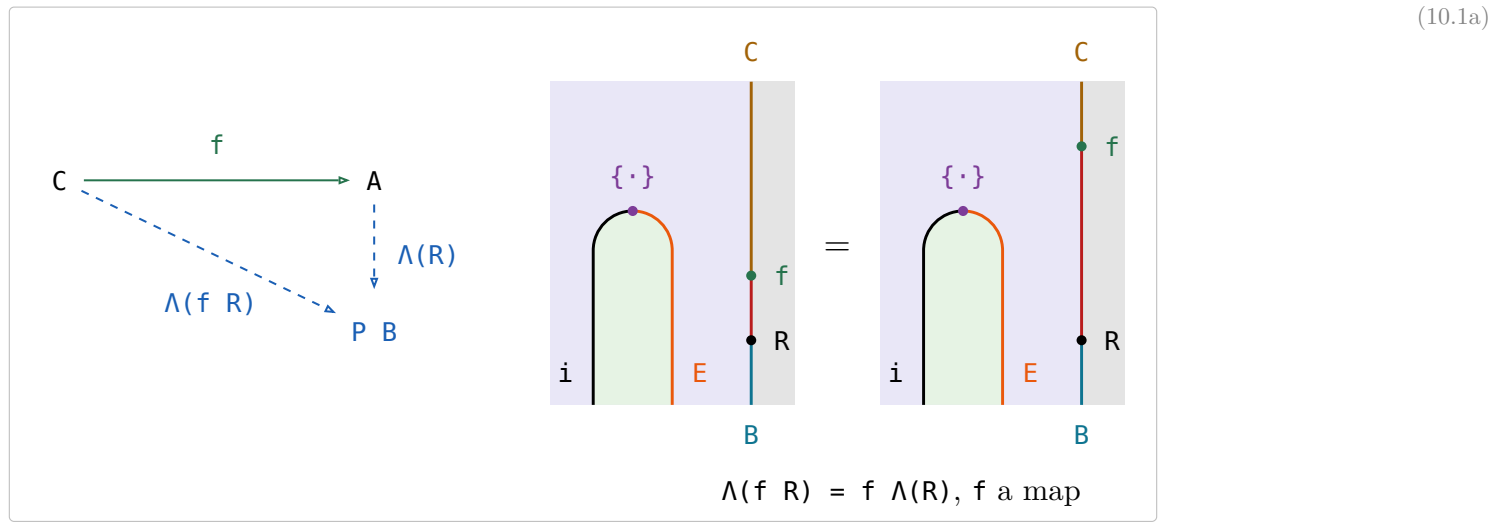
§10. $i \dashv E$

$E(R) \triangleq \Lambda(\exists R)$ is Bird & de Moor's **existential image** (p. 105), and it is the right adjoint of the inclusion $i : \text{Map}(\mathcal{A}) \hookrightarrow \mathcal{A}$: Λ is the transpose, so $\mathcal{A}(A, B) \cong \text{Map}(A, P B)$.

part of $i \dashv E$	the statement	(10a)
$\exists$ the counit, natural	$E(R) \exists = \exists R$	
$\{\cdot\}$ the unit, natural	$f \{\cdot\} = \{\cdot\} E(f)$, for f a map	
Λ the transpose	$\Lambda(R) \exists = R$, and $\Lambda(R)$ the only map with it	
$\cdot \exists$ the transpose back	$\{\cdot\} \exists = 1$ and $\Lambda(\exists) = 1$, the two triangles	



§10.1. Fusion law



(10.1c)

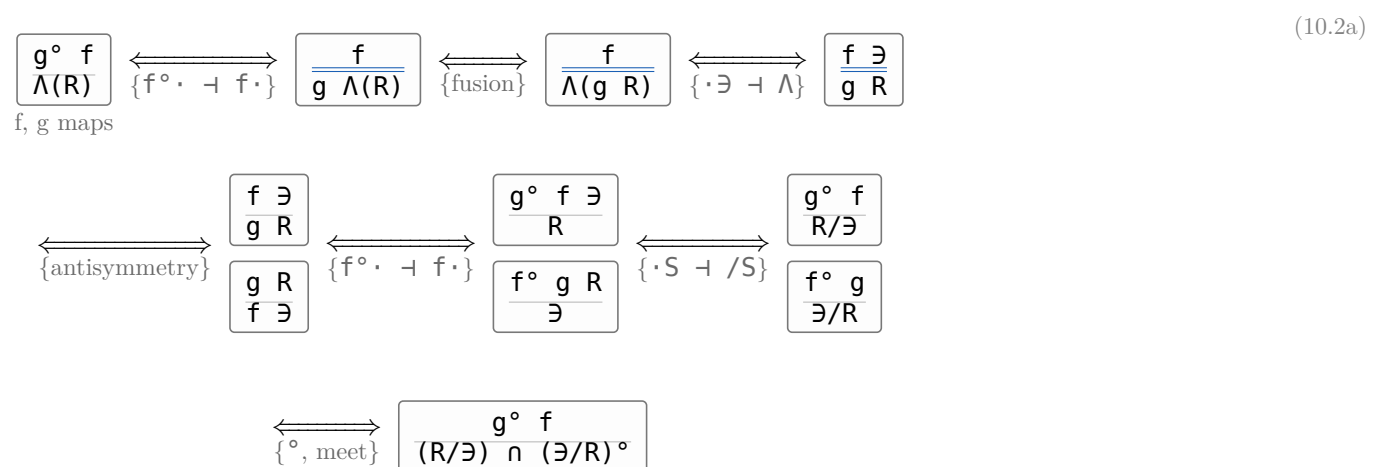
scan line	the object on it	the bead just passed
the top edge	C	—
between $\{\cdot\}$ and S	$i \ E \ C = P \ C$	$\{\cdot\} : C \rightarrow P \ C$
between S and R	$i \ E \ A = P \ A$	S, inside the pair, so $E(S) : P \ C \rightarrow P \ A$
the bottom edge	$i \ E \ B = P \ B$	R, likewise $E(R) : P \ A \rightarrow P \ B$

One string of beads, two groupings:

- The first two, then the third: $\{\cdot\} E(S) = \Lambda(S) : C \rightarrow P \ A$ followed by $E(R) : P \ A \rightarrow P \ B$.
- All three at once: $\{\cdot\} E(S) E(R) = \{\cdot\} E(S R) = \Lambda(S R) : C \rightarrow P \ B$.

Absorption holds for every S, so fusion is its case $S := f$, plus one step: absorption there reads $\Lambda(f R) = \Lambda(f) E(R)$, and turning $\Lambda(f) E(R)$ into $f \Lambda(R)$ needs $\Lambda(f) = f \{\cdot\}$, which is (10.1a) at $R = 1$. That step is the unit's naturality, and it is all fusion adds.

§10.2. $\Lambda(R) = (R/\exists) \cap (\exists/R)^\circ$



§11. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(R \circ S) = F(R) \circ F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

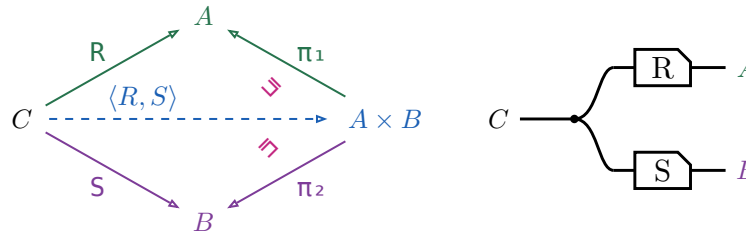
- For f a map, $F(f)$ is a map and $F(f^\circ) = F(f)^\circ$.
- Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$.
- A relator is fixed by what it does to maps.
- $F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly.
- $F(X \cap Y) = F(X) \cap F(Y)$ for X, Y coreflexive.
- $F(\text{Dom } R) = \text{Dom}(F(R))$ for F preserving $\circ$.

The **power relator** $P \multimap x \ P(R) \ y \iff (\forall a \in x. \exists b \in y. a \ R \ b) \wedge (\forall b \in y. \exists a \in x. a \ R \ b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§12. Fork $\langle R, S \rangle$

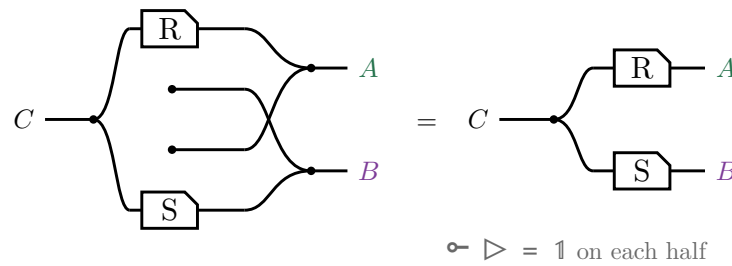
The **fork** of $R : C \rightarrow A$ and $S : C \rightarrow B$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R \quad \langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In $\mathbf{Rel}$, $c \ \langle R, S \rangle \ (a, b)$ iff $c \ R \ a$ and $c \ S \ b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

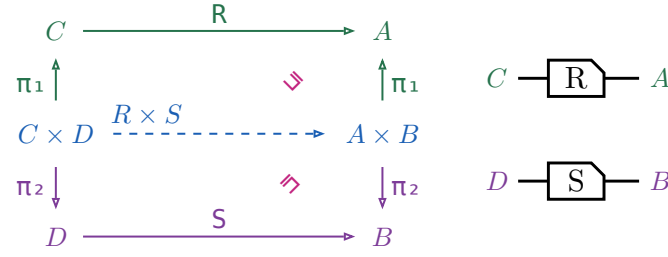
No $\circ$ survives the translation. $\pi_1 = 1 \otimes \multimap$ discards the second component, so $\pi_1^\circ = 1 \otimes \multimap$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§12.1. Relational product $R \times S$

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

(12.1b)

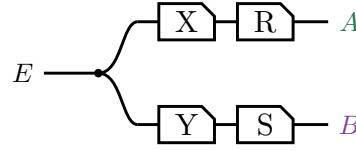


Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§12.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$. Both sides are this picture:

(12.2a)



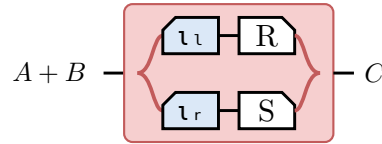
§13. Coproduct $[R, S]$

The injections $\iota_l : A \rightarrow A + B$ and $\iota_r : B \rightarrow A + B$ are maps, and the coproduct they make of the maps stays a coproduct once every arrow is allowed: both equations hold with equality and $[R, S]$ is the only arrow satisfying them, with none of the $\mathbf{Dom}$ slack $\langle R, S \rangle$ carries.

(13a)

$$[R, S] \triangleq \iota_l^\circ R \cup \iota_r^\circ S, \text{ and } R + S \triangleq [R \iota_l, S \iota_r].$$

(13b)



The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.

(13c)

$\iota_l [R, S] = R, \iota_r [R, S] = S, \text{ and } [R, S] \text{ is the only such arrow}$
$\iota_l \iota_l^\circ = 1 = \iota_r \iota_r^\circ$
$\iota_l \iota_r^\circ = \perp = \iota_r \iota_l^\circ$
$\iota_l^\circ \iota_l \cup \iota_r^\circ \iota_r = 1$
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$

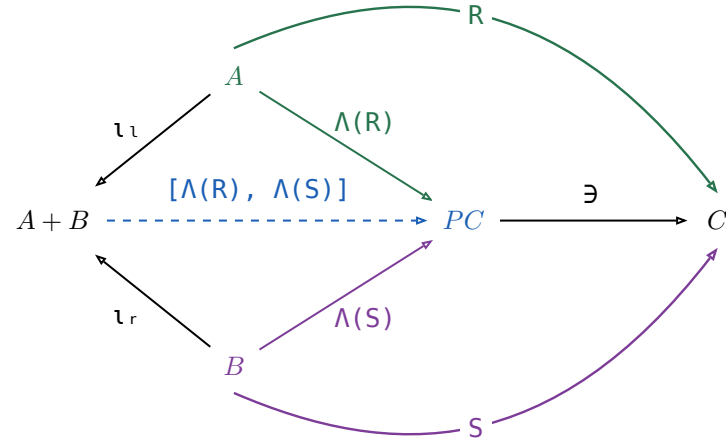
The first row is not free: ι_l, ι_r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because Λ sends an arrow $A \rightarrow C$ to a map $A \rightarrow P \ C$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \rightarrow C$,

(13d)

$$\begin{aligned}
 & \Leftrightarrow \iota_l T = R \text{ and } \iota_r T = S \\
 & \Leftrightarrow \Lambda \text{ is injective — } \Lambda(X) \ni = X \text{ gives } X \text{ back} \\
 & \Leftrightarrow \Lambda(\iota_l T) = \Lambda(R) \text{ and } \Lambda(\iota_r T) = \Lambda(S) \\
 & \Leftrightarrow \Lambda \text{ fusion, } \Lambda(f X) = f \Lambda(X) \text{ for } f \text{ a map} \\
 & \Leftrightarrow \iota_l \Lambda(T) = \Lambda(R) \text{ and } \iota_r \Lambda(T) = \Lambda(S) \\
 & \Leftrightarrow \text{the coproduct of maps, at the map } \Lambda(T) \\
 & \Leftrightarrow \Lambda(T) = [\Lambda(R), \Lambda(S)] \\
 & \Leftrightarrow \Lambda(X) \text{ is the only map with } \Lambda(X) \ni = X \\
 & \Leftrightarrow T = [\Lambda(R), \Lambda(S)] \ni
 \end{aligned}$$

Every step is an $\iff$, so $[\Lambda(R), \Lambda(S)] \ni$ is the **only** arrow satisfying the two equations — which is what $[R, S]$ was claimed to be. Drawn:

(13e)



Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\Lambda(R), \Lambda(S)] \ni$, and pushing $\ni$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\iota_l^\circ \Lambda(R) \cup \iota_r^\circ \Lambda(S)) \ni = \iota_l^\circ R \cup \iota_r^\circ S$.

§14. The power relator $P(R)$

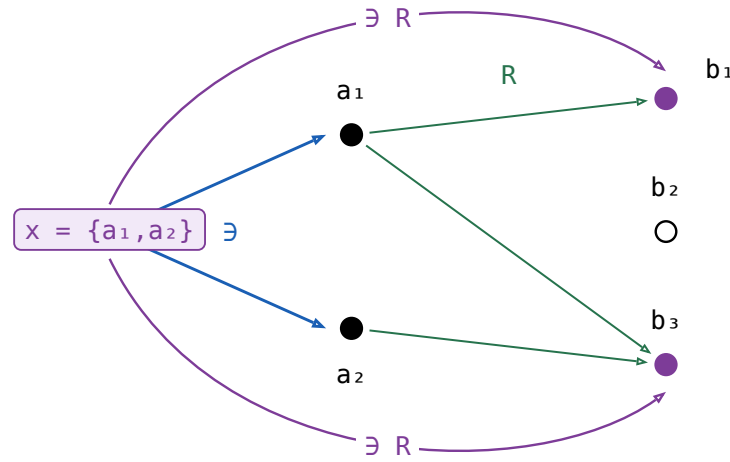
For $R : A \rightarrow B$, $P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : P A \rightarrow P B$.

(14a)

$\exists R$ is a **composition** — $\exists : P A \rightarrow A$ followed by R — and not Freyd's subscripted $\exists_R$, which is the $\exists$ at R 's **target**: §9 writes one $\exists$ per object and drops the subscript. Piece by piece, on one instance, with x , y sets and a , b their elements.

$x = \{a_1, a_2\}$, and R sends a_1 to b_1 and b_3 , a_2 to b_3 alone. Nothing on x reaches b_2 , so b_2 is drawn hollow.

(14b)

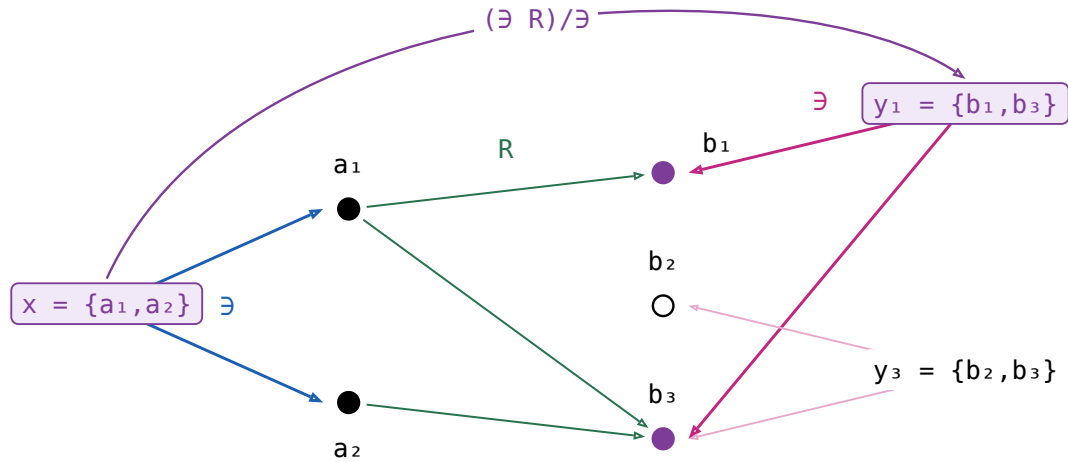


$$\exists R : P A \rightarrow B \quad x (\exists R) b \iff \exists a \in x. a R b$$

one arc per element of B that x reaches, and b_2 gets none

Dividing by $\exists$ turns that into a relation between **sets**: every element of y must be one of the filled dots.

(14c)

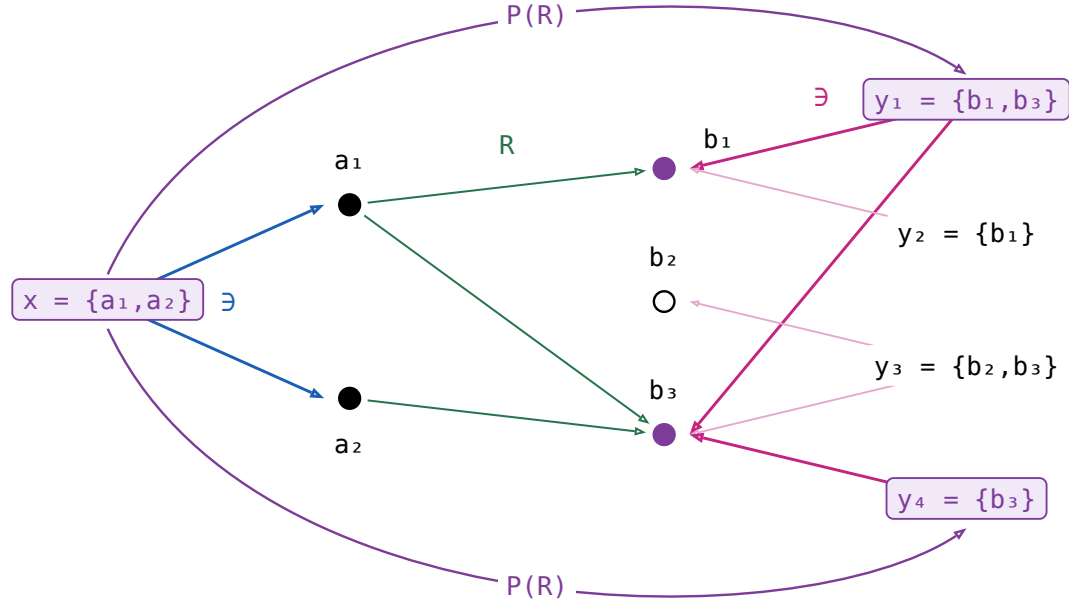


$$(\exists R)/\exists : P A \rightarrow P B \quad x ((\exists R)/\exists) y \iff \forall b \in y. \exists a \in x. a R b$$

y_3 is rejected: it names b_2 , which x does not reach

The other half of the meet is that clause with the sets swapped and R turned round: every element of x must reach y . Both together, and two more sets to separate them:

(14d)



$$((\exists R^\circ)/\exists)^\circ : P A \rightarrow P B \quad x ((\exists R^\circ)/\exists)^\circ y \iff \forall a \in x. \exists b \in y. a R b$$

y_2 is rejected too: a_2 's only image is b_3 , which y_2 does not name

In words, if $x P(R) y$ then every element of x is related by R to some element of y , and conversely — the two clauses of n , one per direction.

Neither $\exists$ nor $P(R)$ is a map. $\exists$ sends a set to **each** of its elements; $P(R)$ sends x to **every** y meeting the two clauses, y_1 and y_4 both. Nor is it entire: an $a \in x$ with no image at all leaves x with no partner whatever. Turning a relation into a map is Λ 's job (§9), and on sets that map is $\Lambda(\exists R)$ — of the four it accepts y_1 only.

Not a symmetric division. $\frac{\exists R}{\exists}$ is $\Lambda(\exists R)$, and in $\mathbf{Rel}$ it reads $x \Lambda(\exists R) y \iff y = \{b \mid \exists a \in x. a R b\}$: in words, y is **exactly** what x reaches, where $P(R)$ asks only that each side cover the other. Nor can it be repaired: in that fraction's second half $(\exists / (\exists R))^\circ$ the R sits in a denominator, so the operation is antitone there, and a relator has to be monotone. $P(R)$ keeps the first half and takes the second half at R° , which puts R back in a numerator. The fraction returns exactly where the two halves agree — at 1 , and at a map.

(14e)

law	the reading
$X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \text{ and } X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(1) = \frac{\exists}{\exists} = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$P(f) = \Lambda(\exists f) = \frac{\exists f}{\exists}$, for f a map	In $\mathbf{Rel}$, $x P(f) y \iff y = \{f a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f a$ on y ; f has just the one image per a , so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R S) = P(R) P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R S)$.

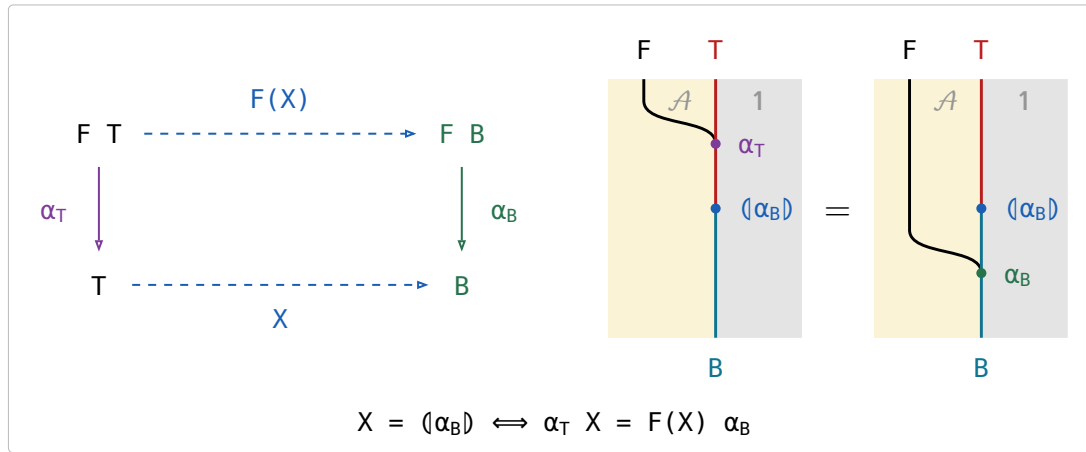
§15. Catamorphism

F a relator with an **initial algebra** $\alpha_T : F\ T \rightarrow T$ among the maps. For a relational algebra $\alpha_B : F\ B \rightarrow B$, the **catamorphism** $(\alpha_B) : T \rightarrow B$ is the unique arrow with $\alpha_T (\alpha_B) = F((\alpha_B)) \alpha_B$.

(15a)

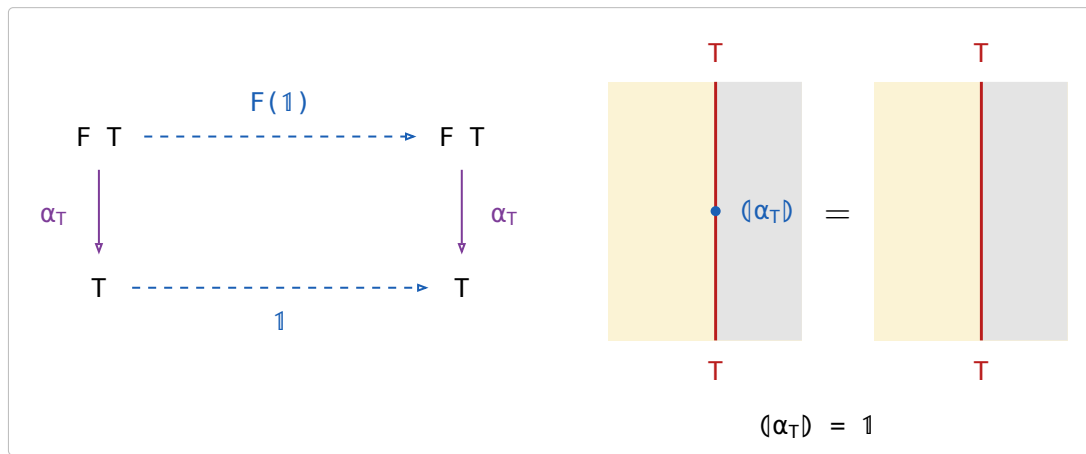
An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $F\ X \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and B .

§15.1. The defining equation



(15.1a)

§15.2. Reflection



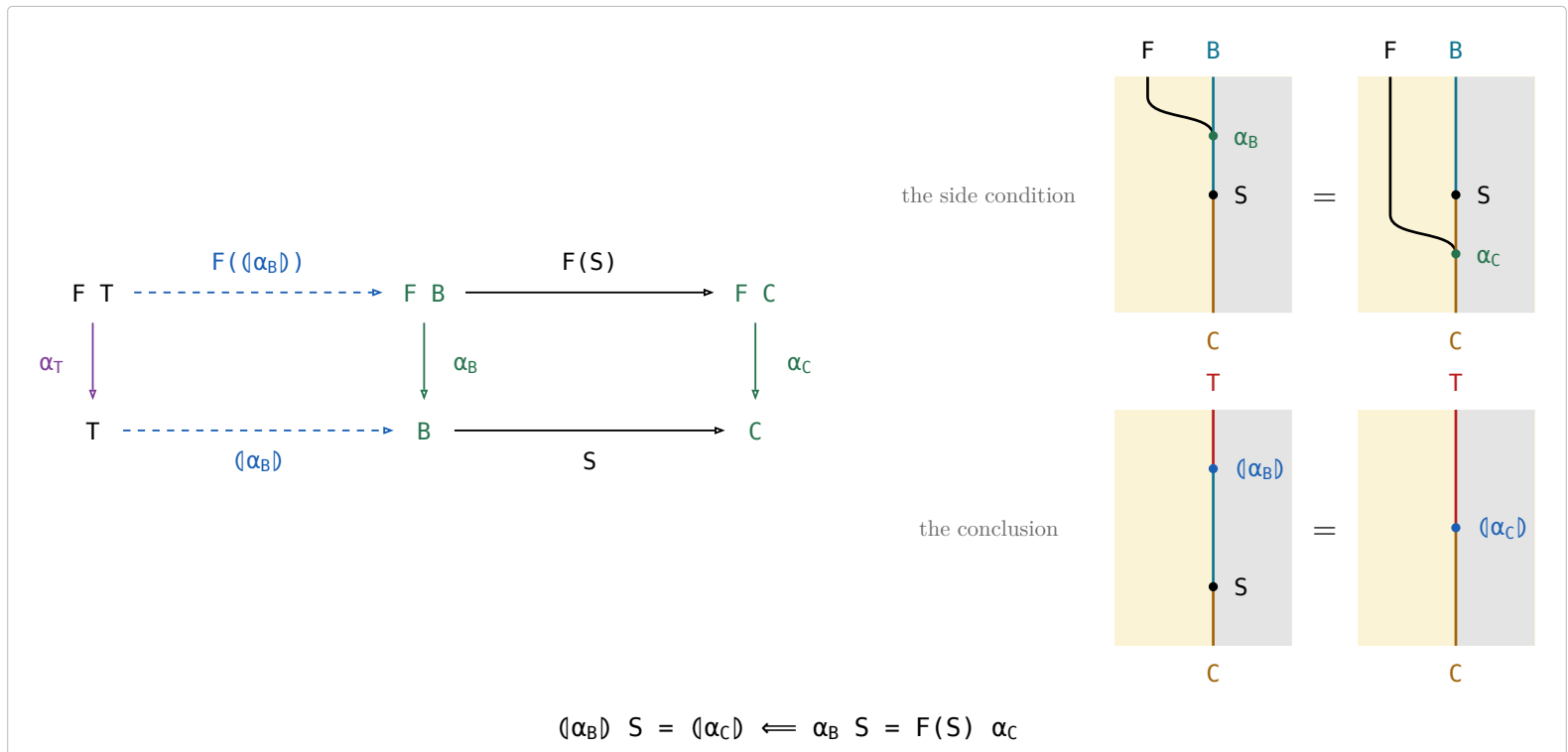
(15.2a)

Taking a value apart with α_T and putting it straight back is doing nothing.

§15.3. Fusion

Fusion rewrites $(\alpha_B) S$ through a second algebra $\alpha_C : F\ C \rightarrow C$ along an arrow $S : B \rightarrow C$.

(15.3a)



The left square is (α_B) 's own defining square and the right one is the side condition, so the outer rectangle says $(\alpha_B) S$ satisfies the defining equation of (α_C) — and uniqueness finishes it. A fold followed by S has collapsed into a single fold, which is how an intermediate structure is got rid of.

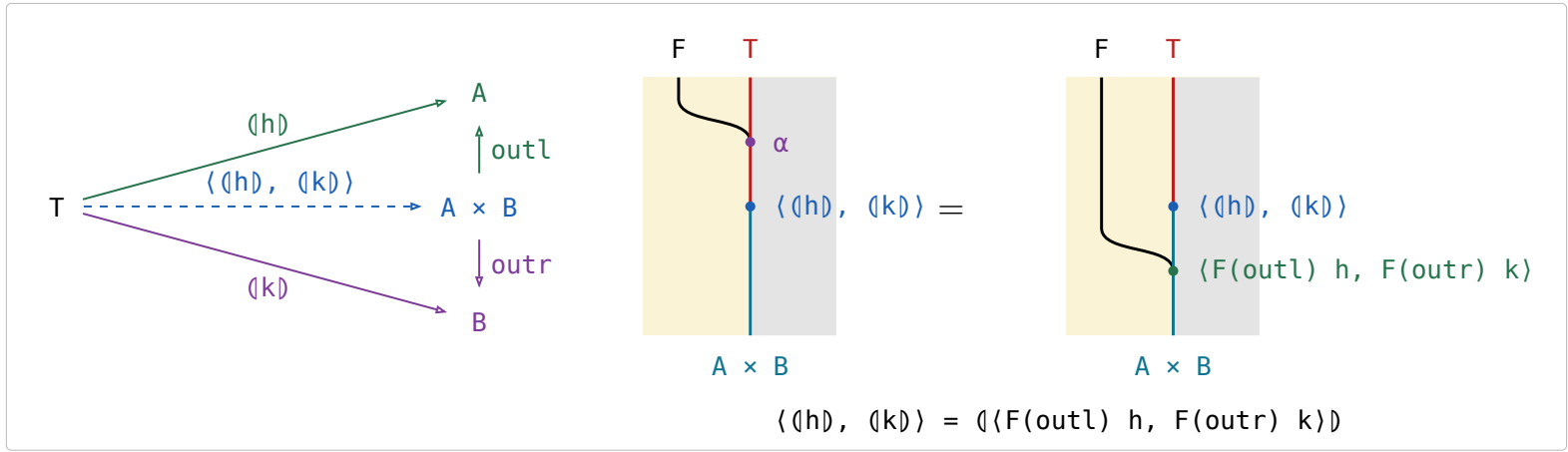
The side condition is (15.1a)'s picture again under $\alpha_T \mapsto \alpha_B$, $(\alpha_B) \mapsto S$, $\alpha_B \mapsto \alpha_C$: both say that the bead between the two merges is a homomorphism of F -algebras.

§16. Catamorphism in $\mathcal{Set}$

(16a)

name	definition	note
listr	<code>listr A ::= nil cons (A, listr A)</code>	The cons-lists over A , the datatype every row below folds (B&dM p. 55).
sum	$\langle [\text{zero}, \text{plus}] \rangle$	$\text{plus}(a, b) = a + b$.
length	$\langle [\text{zero}, \text{outr succ}] \rangle$	<code>outr</code> drops the head and keeps the count of the tail, <code>succ</code> adds one for the head.
average	$\langle \text{sum}, \text{length} \rangle \text{ div}$	$\text{div}(m, n) = m/n$, with $\text{div}(0, 0) = 0$ so <code>average</code> is total. Traverses the list twice.
banana-split law	$\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\text{outl}) h, F(\text{outr}) k \rangle \rangle$	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to	$\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\text{outl}) h, F(\text{outr}) k \rangle$	All that (15.1a) leaves to check: the fork satisfies the defining equation, and uniqueness finishes it.
the instance	$\langle \text{sum}, \text{length} \rangle = \langle [\text{zeros}, \text{pluss}] \rangle$	$\text{zeros} = \langle \text{zero}, \text{zero} \rangle$ and $\text{pluss}(a, (b, n)) = (a + b, n + 1)$, so <code>average</code> runs in one pass.
preds	<code>preds : Nat -> [Nat], preds n = [n, n-1, ..., 1]</code>	B&dM Ex 3.6 (p. 57), posed and not answered there: apply Ex 3.4 to write <code>preds</code> as $\langle k \rangle \text{ outl}$.

(16b)



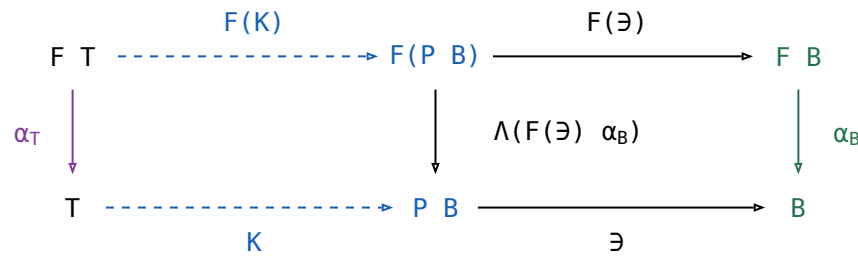
The other laws.

The two $\sqsubseteq$ **fusion** rows rewrite $\langle \alpha_B \rangle\ S$ through the same α_C and S . They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest, the equality fusion (15.3a) included, needs only the initial algebra and the defining equation.

name	law	what it says
Lambek	$\alpha_T^\circ = \alpha_T^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — α_T° takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda(\langle \alpha_B \rangle) = \langle \Lambda(F(\exists)\ \alpha_B) \rangle$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$\langle \alpha_B \rangle = \langle \Lambda(F(\exists)\ \alpha_B) \rangle\ \exists$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$\langle \alpha_C \rangle \sqsubseteq \langle \alpha_B \rangle\ S \iff F(S)\ \alpha_C \sqsubseteq \alpha_B\ S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$\langle \alpha_B \rangle\ S \sqsubseteq \langle \alpha_C \rangle \iff \alpha_B\ S \sqsubseteq F(S)\ \alpha_C$	The other half, with both inclusions turned around.

(16c)

The two Λ rows, drawn:



(16d)

The left square is that catamorphism's own defining square, $K = \langle \Lambda(F(\exists)\ \alpha_B) \rangle$, and the right one is Λ 's cancellation, so the outer rectangle says $K\ \exists$ satisfies the defining equation of $\langle \alpha_B \rangle$ — and uniqueness finishes it.

§16.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ **creates** them: the product of $h : F A \rightarrow A$ and $k : F B \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\text{outl}) h, F(\text{outr}) k \rangle : F(A \times B) \rightarrow A \times B$.
- outl and outr are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (16a) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle F(\text{outl}) h, F(\text{outr}) k \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in (16.1a) the two arrows leave $F(A \times B)$, not $F A$ and $F B$, so each sees BOTH components. The product is the case where they factor as $F(\text{outl}) h$ and $F(\text{outr}) k$; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.

$$\begin{array}{ccc}
 F T & \xrightarrow{F(\langle f, g \rangle)} & F(A \times B) \\
 \alpha \downarrow & & \downarrow h \\
 T & \xrightarrow{f} & A
 \end{array}
 \qquad
 \begin{array}{ccc}
 F T & \xrightarrow{F(\langle f, g \rangle)} & F(A \times B) \\
 \alpha \downarrow & & \downarrow k \\
 T & \xrightarrow{g} & B
 \end{array}$$

$$\alpha f = F(\langle f, g \rangle) h \quad \wedge \quad \alpha g = F(\langle f, g \rangle) k \quad \equiv \quad \langle f, g \rangle = \langle \langle h, k \rangle \rangle$$

(16.1a)

§16.2. Ruby triangles

stage	definition
informally (p. 58)	$\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$
for cons-lists (p. 59)	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$
with the base functor named	$F(A, B) = 1 + A \times B, \alpha = [\text{nil}, \text{cons}], \text{ so } \text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$
abstractly (p. 59)	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$

(16.2a)

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : T A \rightarrow T A$.

$$\begin{array}{ccc}
 T A & \xrightarrow{\text{tri}(f)} & T A \\
 \downarrow \langle F(1, f) g \rangle & & \downarrow \langle g \rangle \\
 & A &
 \end{array}$$

$$\begin{array}{ccc}
 T A & & T A \\
 \downarrow \text{tri}(f) & = & \downarrow \langle F(1, f) g \rangle \\
 A & & A
 \end{array}$$

$$\text{tri}(f) \langle g \rangle = \langle F(1, f) g \rangle \quad \Longleftarrow \quad g f = F(f, f) g$$

(16.2b)

(16.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$ (p. 59).

§16.3. Depth of a tree

the datatype	$\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, base functor $F(A, B) = A + B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$	(16.3a)
the fold	$\langle [g, h] \rangle$ is the unique f with $f(\text{tip } a) = g \ a$ and $f(\text{bin } (x, y)) = h \ (f \ x, f \ y)$	
$\text{tree}(f)$	$\text{tree}(f) = \langle F(f, 1) \ [\text{tip}, \text{bin}] \rangle$; pointwise $\text{tree}(f)(\text{tip } a) = \text{tip } (f \ a)$ and $\text{tree}(f)(\text{bin } (x, y)) = \text{bin } (\text{tree}(f) \ x, \text{tree}(f) \ y)$	
max	$\text{max} = \langle [1, \text{bmax}] \rangle$, where $\text{bmax } (a, b)$ is the larger of a and b	
depths	$\text{depths} = \text{tree}(\text{zero}) \ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor	
depth	$\text{depth} = \text{depths} \ \text{max}$	

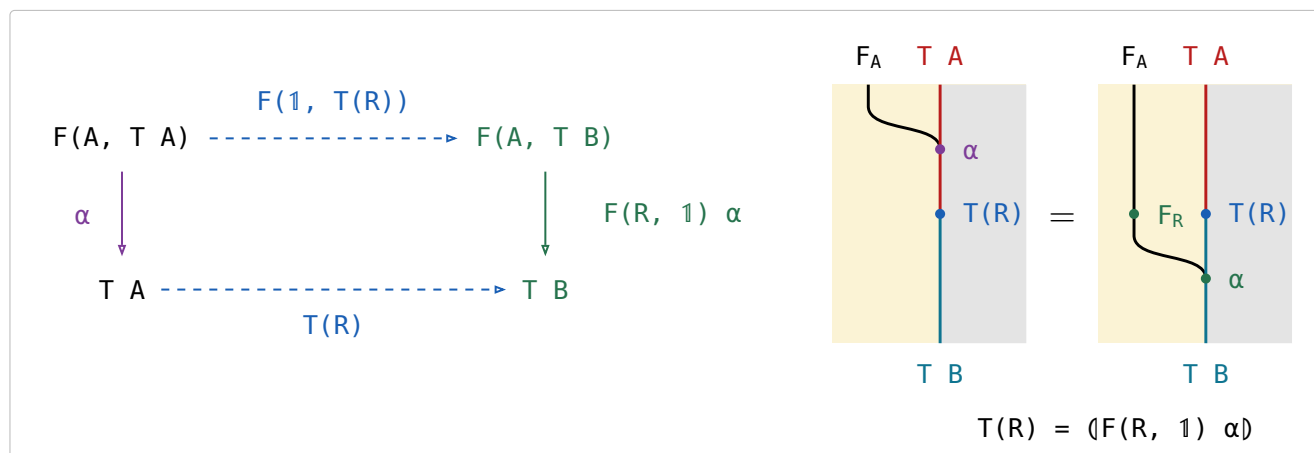
§17. Type functor

F a **binary** relator: $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the catamorphism section. For every object A the initial algebra is $\alpha : F(A, T A) \rightarrow T A$, among the maps. The **type functor** T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha) : T A \rightarrow T B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, $\text{list}(R) = ([\text{nil}, (R \otimes 1) \text{cons}])$, which is $\text{map}(R)$.

A bifunctor puts its two arguments in two different places. The wire is the partial application $F_A = F(A, -)$, and an arrow in the SECOND argument is a bead on the object wire with the F wire running past it — the catamorphism section's rule, unchanged — while an arrow $R : A \rightarrow B$ in the FIRST induces a natural transformation $F_R : F_A \Rightarrow F_B$, which is a bead ON the F wire. Marsden draws bifunctors exactly so ([CatString.pdf](#), **Bifunctors**): side-by-side is already spent on functor composition, so one argument has to go into the wire's name.



$T(R)$ is the unique arrow making it commute; there is no Ξ in it.

Two beads at one height on two different wires are one action, here $F(R, T(R))$, so the right panel reads $F(R, T(R)) \alpha$ and the left one $\alpha T(R)$. Their relative height says nothing: slide F_R below the $T(R)$ bead and the panel reads $F(1, T(R)) F(R, 1)$ and then α , which is the square's top row followed by its right vertical; slide it above and the panel reads $F(R, 1) F(1, T(R))$, the route through $F(B, T A)$ the square leaves out. Identifying those two is the exchange condition every bifunctor satisfies (Hinze & Marsden, Ex 1.23), and in this calculus there is nothing to identify.

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way; that is $\text{map}(R)$.
functor	$T(1) = 1$ and $T(R) T(S) = T(R S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$T(R) (Q) = (F(R, 1) Q)$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha T(R) = F(R, T(R)) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion (15.3a) applied to $T(R)$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R, 1) F(1, (Q)) = F(R, (Q)) = F(1, (Q)) F(R, 1)$ — so it too needs only the initial algebra and the defining equation.

(17d)

$$\begin{array}{ccc}
 F(A, T A) & \xrightarrow{F(R, T(R))} & F(B, T B) \\
 \downarrow \alpha & & \downarrow \alpha \\
 T A & \xrightarrow{T(R)} & T B
 \end{array}$$

It commutes strictly, and it is (17b) with the right vertical's two halves separated: push $F(R, 1)$ up into the top row, where it meets $F(1, T(R))$ as the one action $F(R, T(R))$, and α is left as the vertical. In the string diagram that is the F_R bead sliding up past $T(R)$, and what stays put is α , one bead falling past $T(R)$ from the row above to the row below — which is what “ α is natural” means.

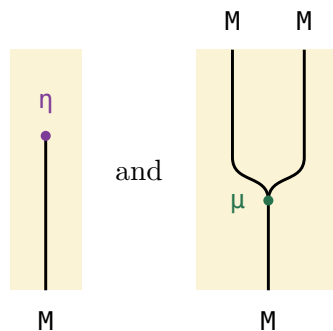
§18. Monad

(18a)

An endofunctor $M : \mathcal{A} \rightarrow \mathcal{A}$ with two natural transformations, the **unit** $\eta : \text{Id} \Rightarrow M$ and the **multiplication** $\mu : M \circ M \Rightarrow M$, subject to a unit law and an associativity law:

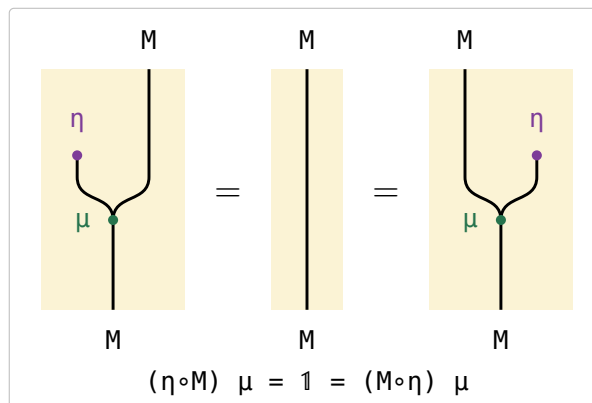
$$(\eta \circ M) \mu = 1 = (M \circ \eta) \mu \qquad (\mu \circ M) \mu = (M \circ \mu) \mu$$

(18b)



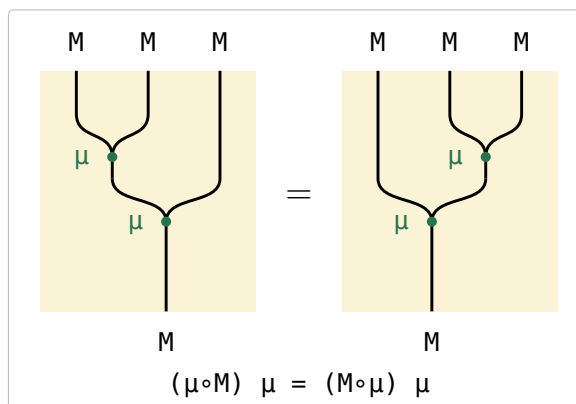
In (18b) the identity functor is a REGION and not a wire, so $\eta : \text{Id} \Rightarrow M$ has nothing above it and the M wire simply begins at its bead — the book's lollipop. $\mu : M \circ M \Rightarrow M$ is the two M wires of $M \circ M$ merging into one, its tuning fork.

(18c)



The lollipop is absorbed into the fork from either side and a bare wire is left, which is what 1 is in this calculus: nothing is drawn for it.

(18d)



The two nestings of the fork agree, so three M s collapse to one whichever pair is multiplied first.

§18.1. Monad in $\mathcal{S}et$

monad	η	μ	η on elements	μ on elements
$\text{Id } A = A$	1	1	$\eta A a = a$	$\mu A a = a$
$\text{Exc } A = A + E$	ι_l	$1 \nabla \iota_r$	$\eta A a = \iota_l a$	$\mu A (\iota_l y) = y$ $\mu A (\iota_r e) = \iota_r e$
$\text{State } A = (A \times P)^P$	$\text{curry } 1$	$(\text{apply}_{A \times P})^P$	$\eta A a = \lambda x . (a, x)$	$\mu A f = \lambda x . g y \text{ where } (g, y) = f x$
$\text{Pow } A = P A$	$\{\cdot\}$	$\bigcup$	$\eta A a = \{a\}$	$\mu A Xs = \bigcup Xs$

(18.1a)

§18.2. The state monad

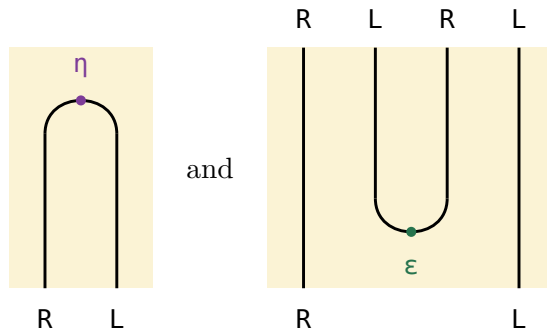
For a fixed set P of states, `IntroString p. 67` defines

$$\text{State } A = (A \times P)^P \quad \eta A = \text{curry } (1 : A \times P \rightarrow A \times P) \quad \mu A = (\text{apply}_{A \times P})^P$$

Both arrows come from the curry adjunction $L \dashv R$, $L A = A \times P$ and $R B = B^P$ (`IntroString p. 96`):

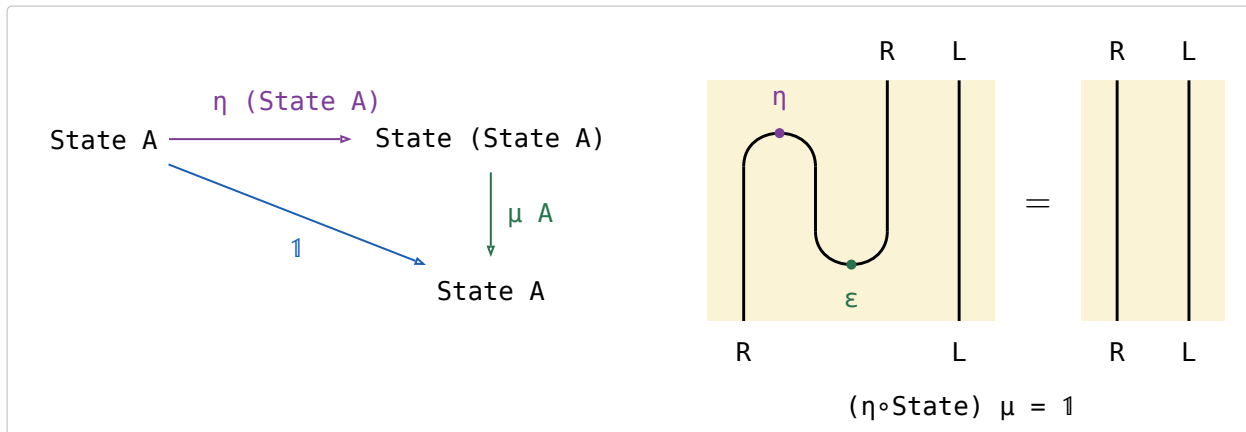
- $\text{State} = R \circ L$.
- η is the unit $\text{Id} \Rightarrow R \circ L$, which is why it is the curried identity.
- the counit $\varepsilon : L \circ R \Rightarrow \text{Id}$ is `apply`.
- $\mu = R \circ \varepsilon \circ L$, that counit with a functor on each side: in $\mu A = (\text{apply}_{A \times P})^P$ the subscript $A \times P$ is $L A$, the functor INSIDE, and the outer $(-)^P$ is R , the functor OUTSIDE.
- $\dashv$ composes nothing; it only names the left adjoint first.

Applicative order stands R left and L right in $\text{State} = R \circ L$. The identity functor is a region, so a unit $\text{Id} \Rightarrow R \circ L$ has nothing above it and two wires below — one strand that OPENS the pair; a counit $L \circ R \Rightarrow \text{Id}$ is that strand upside down, CLOSING a pair that arrives from above.



(18.2a)

In (18.2a), $\text{State} \circ \text{State}$ is the four wires $R L R L$, and $\mu = R \circ \varepsilon \circ L$ closes the middle pair — the $L \circ R$ — leaving the outer R and L , which is State again. The two wires that run past the turn untouched are the two functors composed on, R outside and L inside.



(18.2b)

(18.2b) is the zig-zag. $\eta \circ \text{State}$ opens a new $R L$ left of the pair already there and μ closes the middle two — the new L against the old R — leaving the new R and the old L , which is State unchanged. Strip the old L , which never moves, and what zig-zags is the curry adjunction's snake equation $(\eta \circ R)(R \circ \varepsilon) = 1$ (`IntroString p. 93`): (18c) read at State is that triangle identity with L composed on the inside.

In `Set` it is beta-reduction. $\eta (\text{State } A) f = \lambda x . (f, x)$ pairs the computation with the state, and μA applies it back, $\lambda x . f x$, which is f .